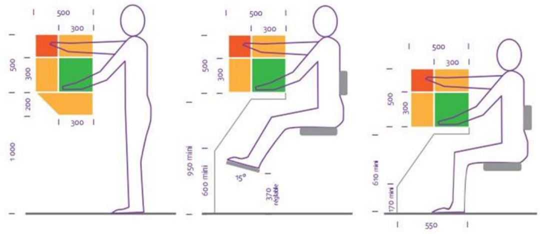

WORKSTATION STUDY
Ergonomic and Risk analysis

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Cursus : IO EENG3

UE :

Version :

Date :

Background **story**

After several work stoppages by an operator on an automobile assembly line.

You need to study the reason for these work stoppages in order to anticipate for the workstation design.

Source: https://www.nsc.org/work-safety/tools-resources/infographics/workplace-injuries

Introduction to the workplace ergonomics

Source:
https://youtu.be/QeDUCXfzI6U

Course goals

- Understand the main principles of ergonomics for workstation design
  - Interpret standard ergonomics data
    - Conduct a Failure Mode and Effects Criticality Analysis (FMECA) to anticipate potential machine failures

Summary

1. Design and improvement of workstation ergonomics
2. The ergonomic standard
3. The ergonomic study
4. Failure Modes Effects and Criticality Analysis

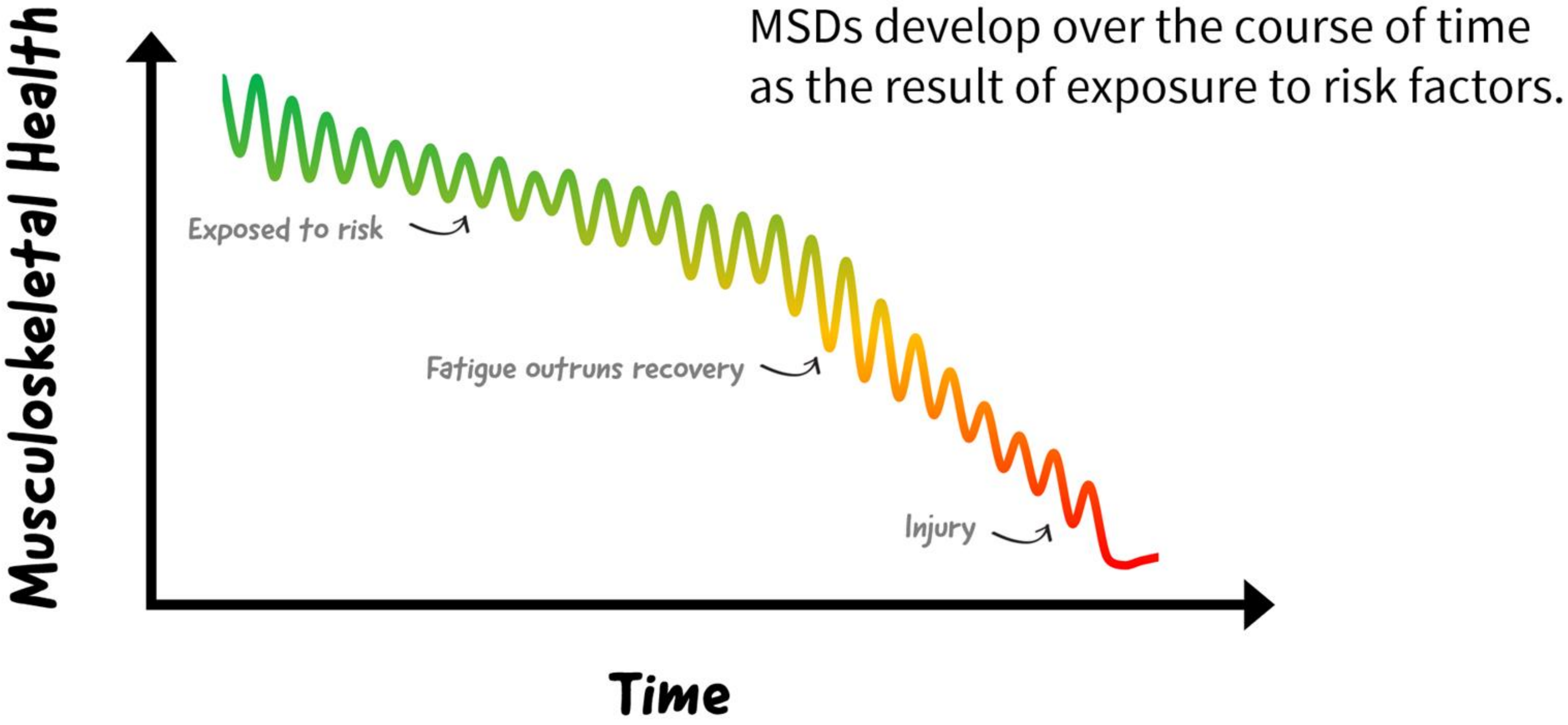
1. Design and improvement of workstation ergonomics

1.Design and improvement of workstation ergonomics

Good to know

Source: https://www.travelers.com/resources/business-industries/manufacturing/top-causes-of-manufacturing-injuries

1.Design and improvement of workstation ergonomics



Musculoskeletal Disorders (MSDs)

1.Design and improvement of workstation ergonomics

Recognize and report ergonomic risks

Force	Posture	Repetition
	 	
Excessive force = Risk!	Awkward posture= Risk!	High repetition = Risk!

Source: https://ergo-plus.com/ergonomics-awareness-training-for-employees-pdf/#download

1.Design and improvement of workstation ergonomics

REMINDER: The 22 rules of economy of movements

Simultaneity of movements

1. Both hands must start and end their movements at the same time
2. Both hands must not remain inactive at the same time, except during rest
3. The movements of the arms must be symmetrical and simultaneous.

Minimum energy expenditure

1. The movements necessary for work must use the smallest possible muscle mass.
2. Continuous movements are preferable to “Zig Zag” movements or movements in broken lines with acute angles

The living force

1. Live force should be used whenever possible to assist the operator's movements. It should be minimized if movement is controlled.
2. Ballistic movements are faster, easier and more precise than constrained or controlled movements.

1.Design and improvement of workstation ergonomics

REMINDER: The 22 rules of economy of movements

Pace

8. Acquiring a pace is essential to the easy and automatic execution of a Job

Order of the equipped work area

8. There must be a defined place for all materials or components.
9. Tools, materials and testers should be placed as close as possible and as far away from the operator as possible.
10. The materials, components and tools must be arranged to allow the best possible sequence of movements.

Use of gravity

8. Gravity feed boxes and containers must supply the performer(s) close to their workplace.
9. Gravity must be used for evacuation: chutes, conveyors, inclined rollers, etc.

1.Design and improvement of workstation ergonomics

REMINDER: The 22 rules of economy of movements

Comfort and lighting of the work surface

14. Each operator must be provided with the best conditions for the lighting of his work
15. The height of the work surface and the seat should as far as possible allow working standing or sitting
16. A seat allowing good posture must be provided to each operator

Freedom of hands

14. The hands must be relieved of all the work which can be done more advantageously by an assembly

Combining / positioning

14. Tools should be combined whenever possible
15. Tools and devices should be positioned whenever possible

Finger loading

14. When each finger performs a separate movement, the load of each finger must be distributed according to the capacities of each

1.Design and improvement of workstation ergonomics

REMINDER: The 22 rules of economy of movements

Control elements

21. The handles must allow the largest possible contact surface
22. The levers and flywheels must allow them to be maneuvered with the slightest change in posture and with as much mechanical efficiency as possible.

2.The ergonomic standard

2.The ergonomic standard

The ergonomics standard, co-construction for the improvement of working conditions and the prevention of arduous work.

space
means
signage
representation
orders

Based on:

- *NF / ISO / EN Standards*
- *The legislation*

lighting
thermal
sound
vibration
chemical

The ergonomic standard

morphology
postures
efforts
movements

versatility
multi-skill
rotation
breaks
capabilities

Enriched by experience allowing weighting of the normative elements which are recommendations

2.The ergonomic standard

Useful links

OSHA : owww.osha.gov
- Occupational Safety & Health Administration - US department of labor

OIT : www.ilo.org
- Organisation Internationale du Travail - Ergonomie

UCLIC : www.ergohci.ucl.ac.uk
- University college London Interaction Center

OEHS : www.keats.admin.virginia.edu
- Office of environemental health & safety - University of Virginia

CEN : www.cenorm.be
- Comité Europeen de la normalisation - European comittee for standardization

ISO : www.iso.org
- International Organization for Standardization - GB / FR

NIESH : www.niesh.nih.com
- National Institute of Environmental Health Science

CCOHS : www.ccohs.ca
- Canadian Center for Occupationnal Health & Safety - GB / FR

HFESA : www.ergonomics.org.au
- Human Factor & Ergonomics Society of Australia

SELF : www.ergonomie-self.org
- Société d'ergonomie de Langue Française

CUERGO : www.ergo.human.cornell.edu
- Cornell University Ergonomics

IEA : www.iea.cc
- International Ergonomic Association

AFNOR : www.afnor.fr
- Association Française de Normalisation

INRS : www.inrs.fr
- Institut National sur la Recherche et la Sécurité

NIOSH : www.cdc.gov/niosh/
- Center for Disease Control and Prevention - National Institute of Occupational Health

Washington State Department of Labor & Industries : www.lni.wa.gov

The Ergonomics Society : www.ergonomics.org.uk

3.The ergonomic study

3.The ergonomic study

8 Fundamental Ergonomic Principles for better work performance:

- Principle 1. Maintain Neutral Posture
- Principle 2. Work in the Power / Comfort Zone
- Principle 3. Allow for Movement and Stretching
- Principle 4. Reduce Excessive Force
- Principle 5. Reduce Excessive Motions
- Principle 6. Minimize Contact Stress
- Principle 7. Reduce Excessive Vibration
- Principle 8. Provide Adequate Lighting

https://ergo-plus.com/fundamental-ergonomic-principles/

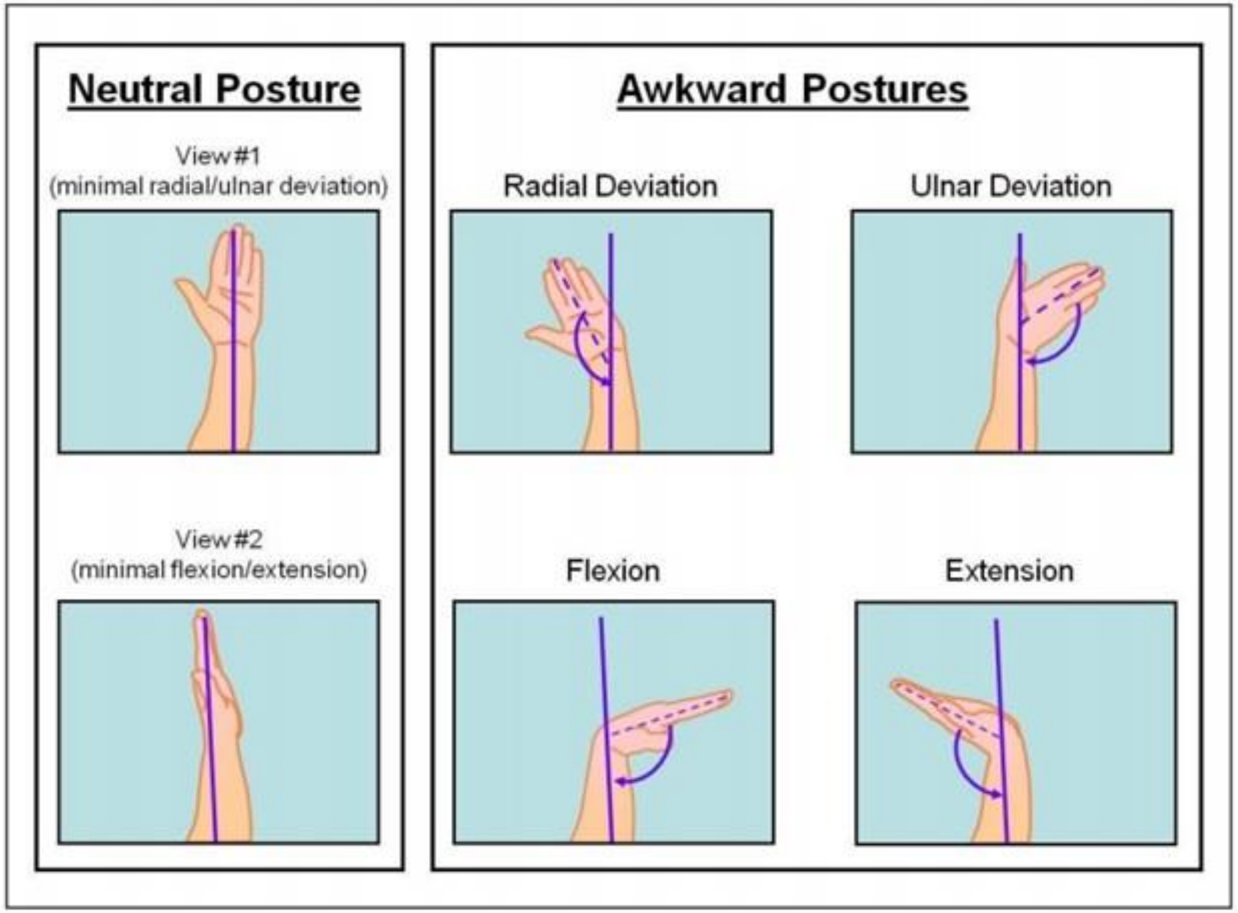
3.The ergonomic study

Principle 1. Maintain Neutral Posture

- **Neutral postures** are postures where the body is aligned and balanced while either sitting or standing, placing minimal stress on the body and keeping joints aligned.

- The opposite of a neutral posture is an "**awkward posture**." Awkward postures move away from the neutral posture toward the extremes in range of motion. This puts more stress on the worker's musculoskeletal system, is a contributing risk factor for [Musculoskeletal Disorders (MSDs)](), and should be avoided.

Neutral and awkward wrist postures

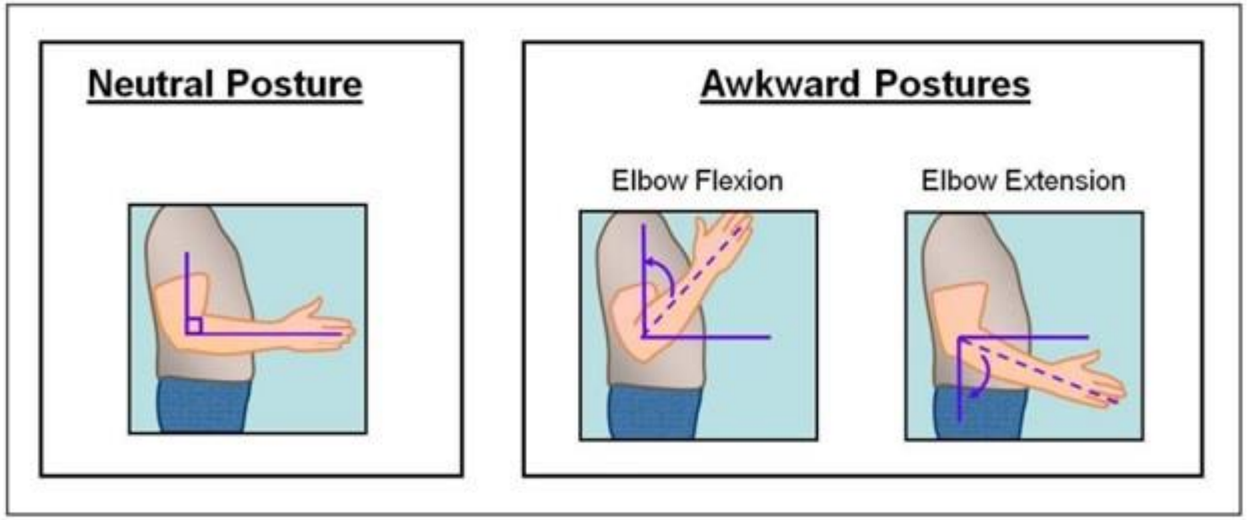


https://ergo-plus.com/fundamental-ergonomic-principles/

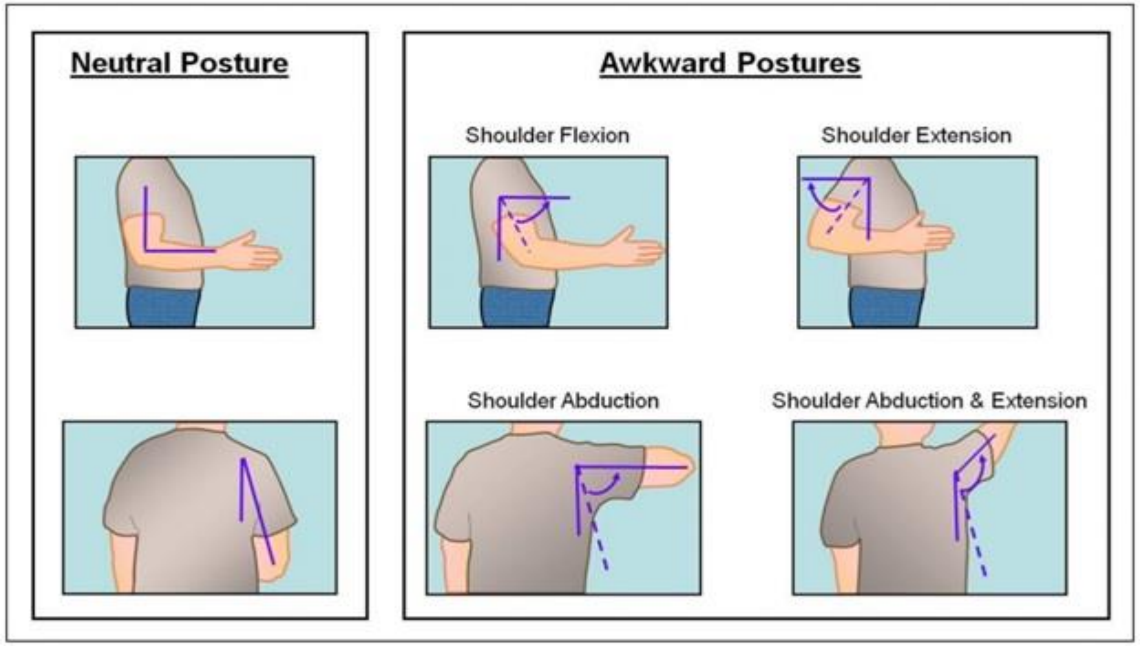
3.The ergonomic study

Principle 1. Maintain Neutral Posture

Neutral and awkward elbow postures



Neutral and awkward shoulder postures



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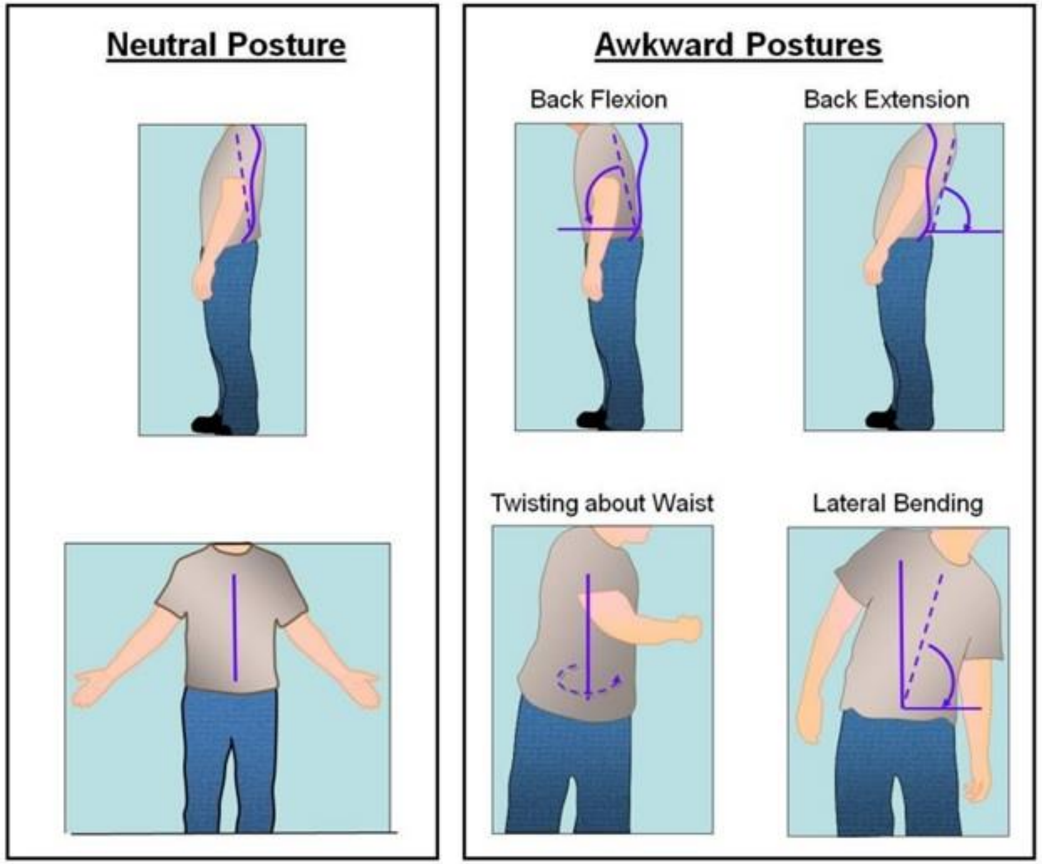
Pistol grip vs. inline grip drivers to maintain neutral posture

Principle 1. Maintain Neutral Posture

Neutral and awkward back postures



Bent Wrist → Poor Wrist Position

Straight Wrist → Ideal Wrist Position

Bent Wrist → Poor Wrist Position

Straight Wrist → Ideal Wrist Position

3.The ergonomic **study**

Principle 1. Maintain Neutral Posture > Choice of Work Posture

Posture is the element to be defined first, the other elements of the workstation are to be defined in relation to it.

The most common cause of fatigue is due to the work posture imposed by the design of the position.

Workstation

Work done in areas of arm reach

Work done on a large horizontal / vertical surface or work with bulky objects

Low force requirement for handling light objects

High force requirement for handling heavy objects

Sufficient legroom

Insufficient legroom

Special requirements for working height

Standing with support

Standing

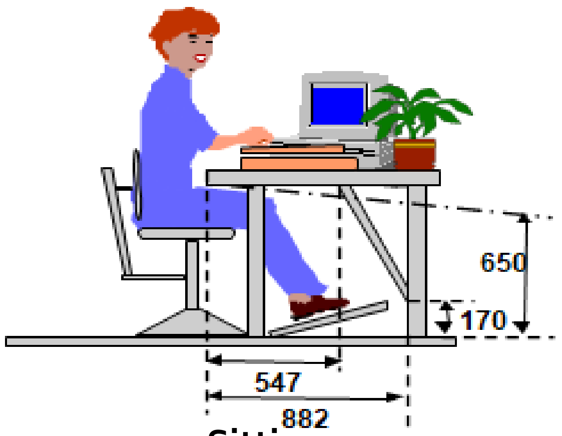
Sitting

Sitting raised

Posture to remember in the case of precision work

3.The ergonomic study

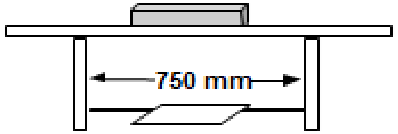
Principle 1. Maintain Neutral Posture > Choice of Work Posture

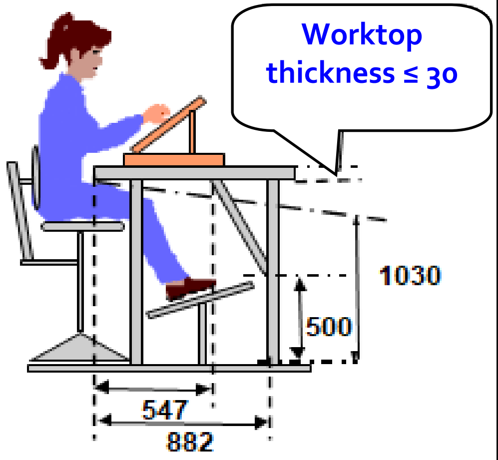


Sitting post

Clearance under the Workstation

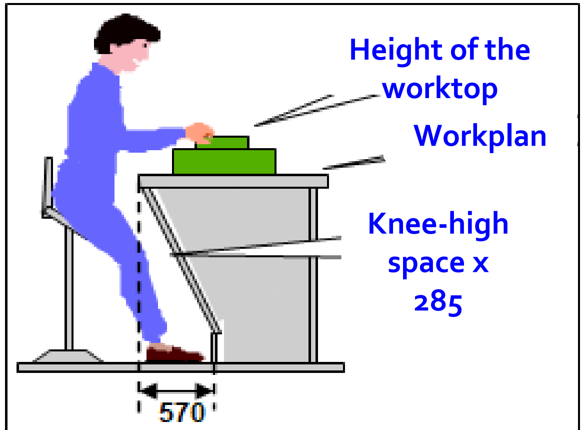
Minimum clearance to be respected for mobility of the lower limbs



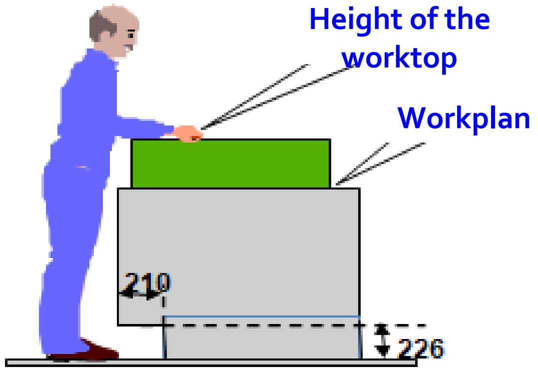


Seated-Elevated Post

The lateral clearance width under the seated workstation for leg passage must be at least 750 mm



Standing position with support

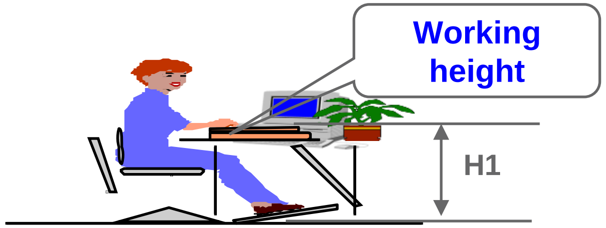
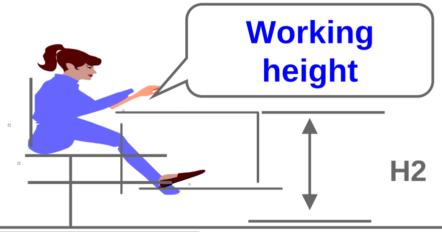
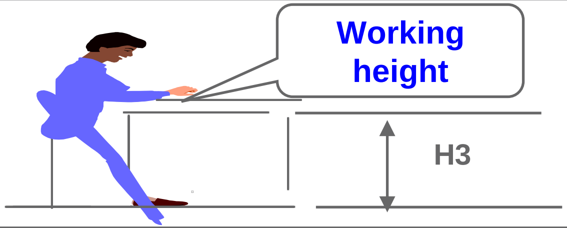
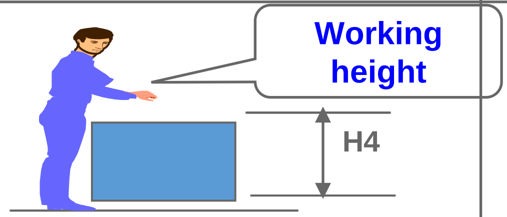


Standing position

3.The ergonomic study

Principle 1. Maintain Neutral Posture > Height and Work Plan

Choice table

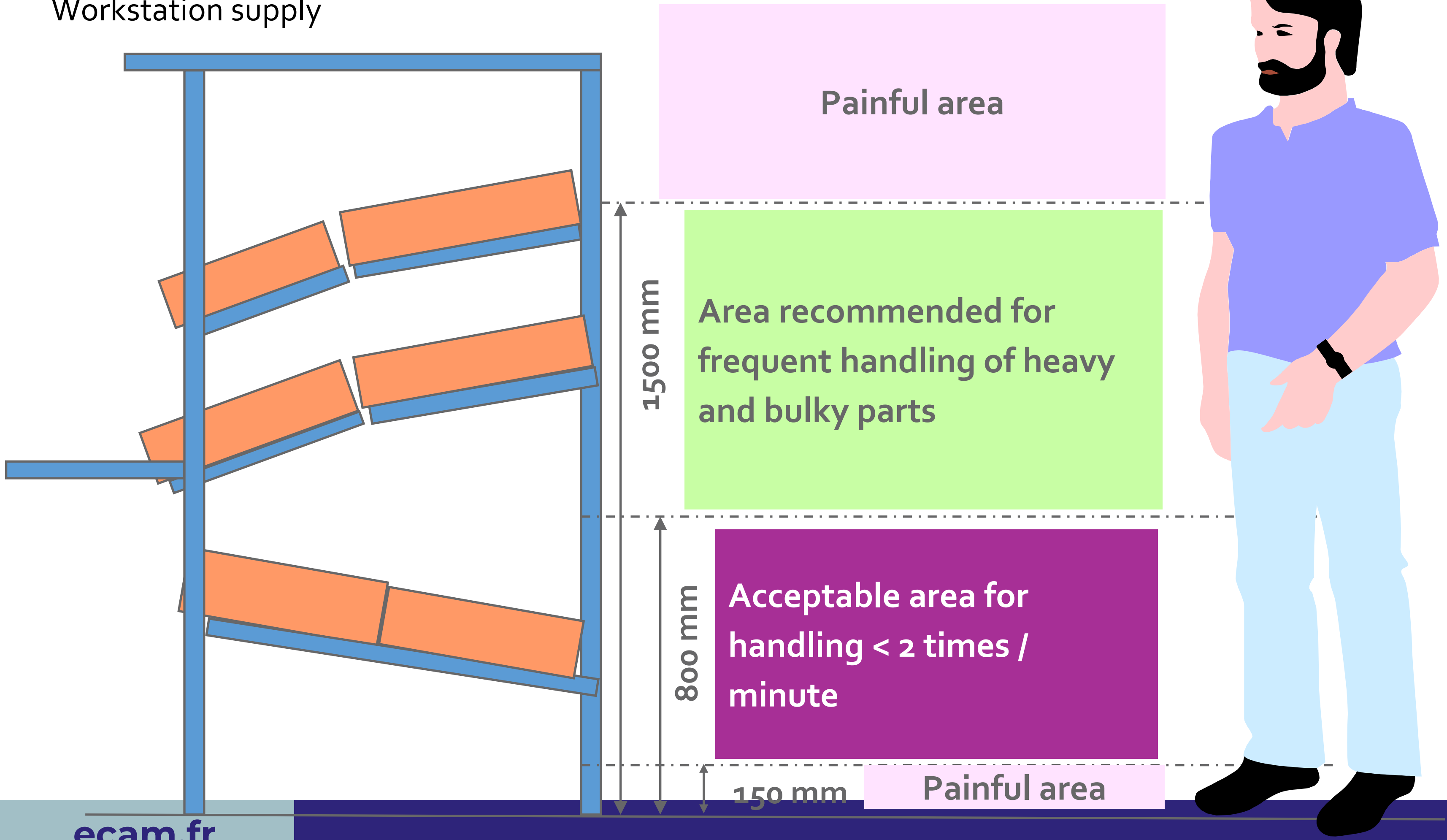
Task requirements	Position	Working height	Height of the worktop
Office work		H1 = 650 à 750 mm	Worktop at elbow height
High vision and precision		H2 = 1100 à 1200 mm	Possibility of a high work plan
Average vision and accuracy		H3 = 1000 à 1100 mm	Worktop at elbow height
Handling bulky and / or heavy objects		H4 = 800 à 1000 mm	Worktop lower than the elbow height

3.The ergonomic study

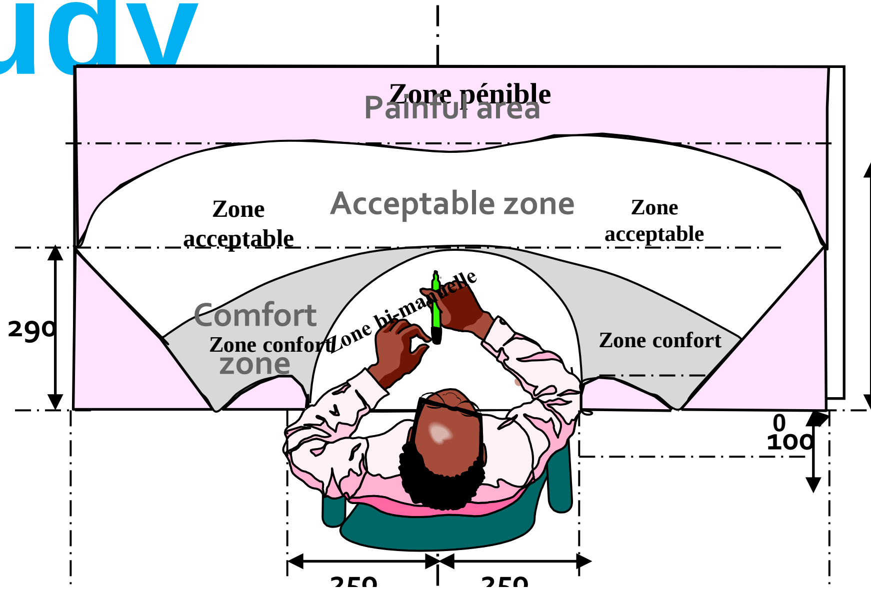
https://www.osha.gov/

Principle 2. Work in the Power / Comfort Zone

Workstation supply



3.The ergonomic study



Principle 3. **Allow for Movement and Stretching**

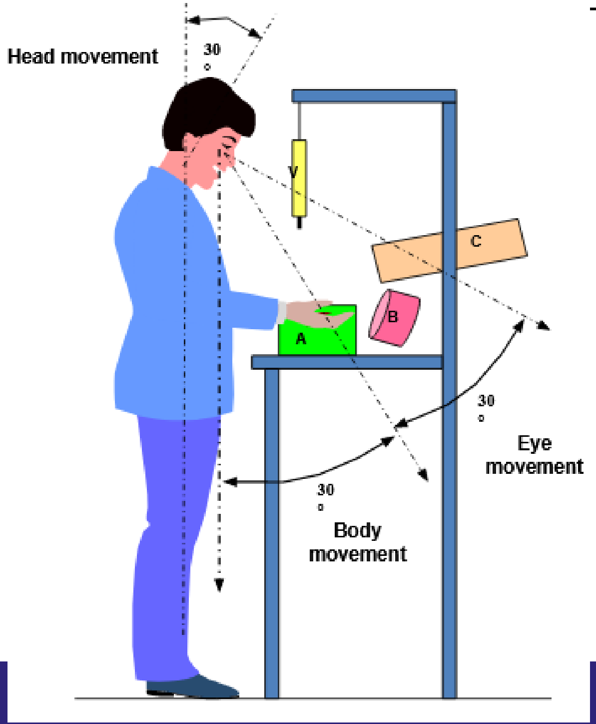
Define the position of the workpieces or tools to be handled according to their use by respecting the prescribed zones.

Principle 4. **Reduce Excessive Force**

Reminder: The reduction of risks requires the provision of handling aids and/or the lightening of loads.

Recommendations :

- Bulky and heavy objects > 12kg
- Cartons, unit containers < 12 kg
- Taking parts in containers < 12 kg
- Use roller conveyors to prevent products or cartons from getting crooked, snagging and making handling difficult.

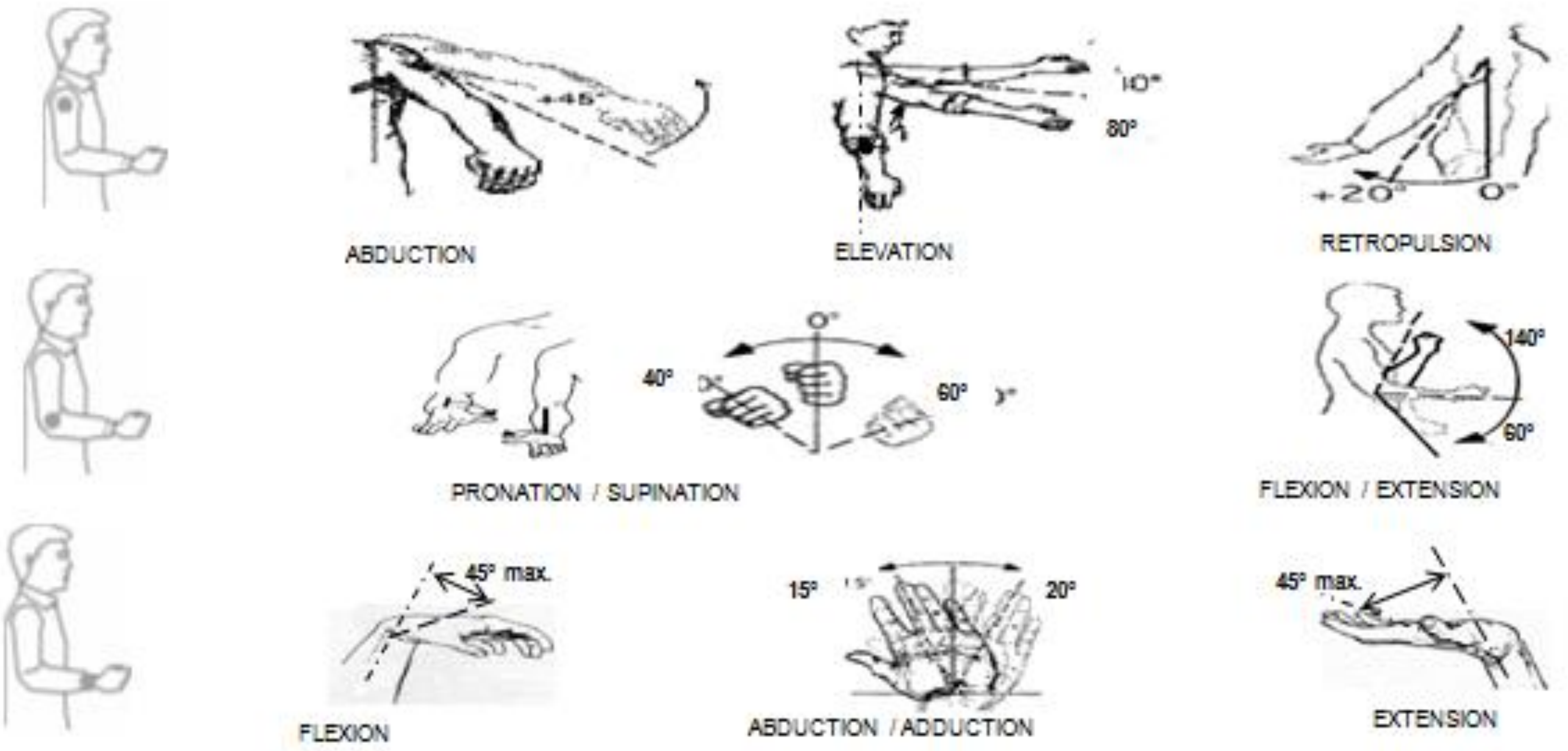


(source)

3.The ergonomic study

Principle 5. Reduce Excessive Motions > **Recommendation for handling actions**

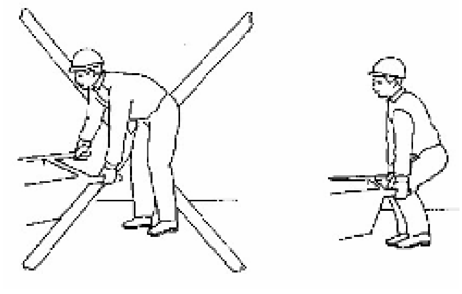
When defining operating procedures, try to limit extreme angulations and repetitive gestures. Take into account the forces exerted and the combinations of gestures.

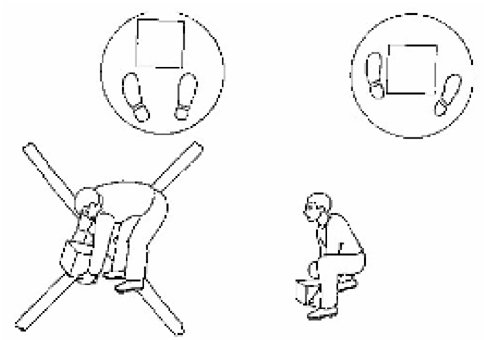


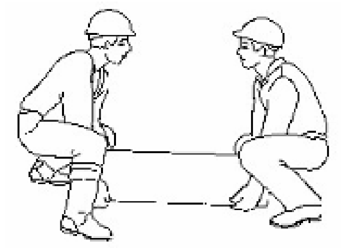
Repetitive gestures

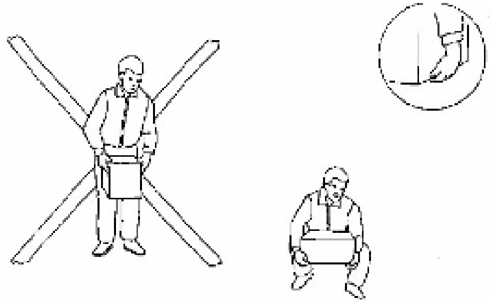
3.The ergonomic study

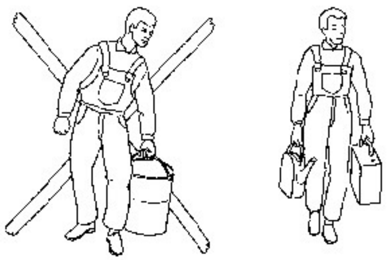
Principle 5. Reduce Excessive Motions > Recommendation for handling actions

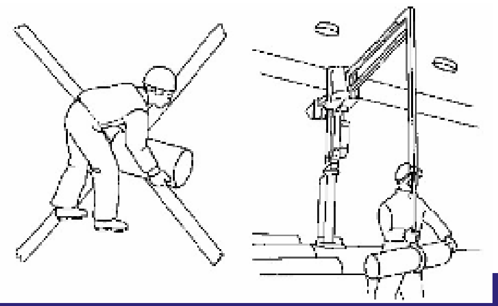












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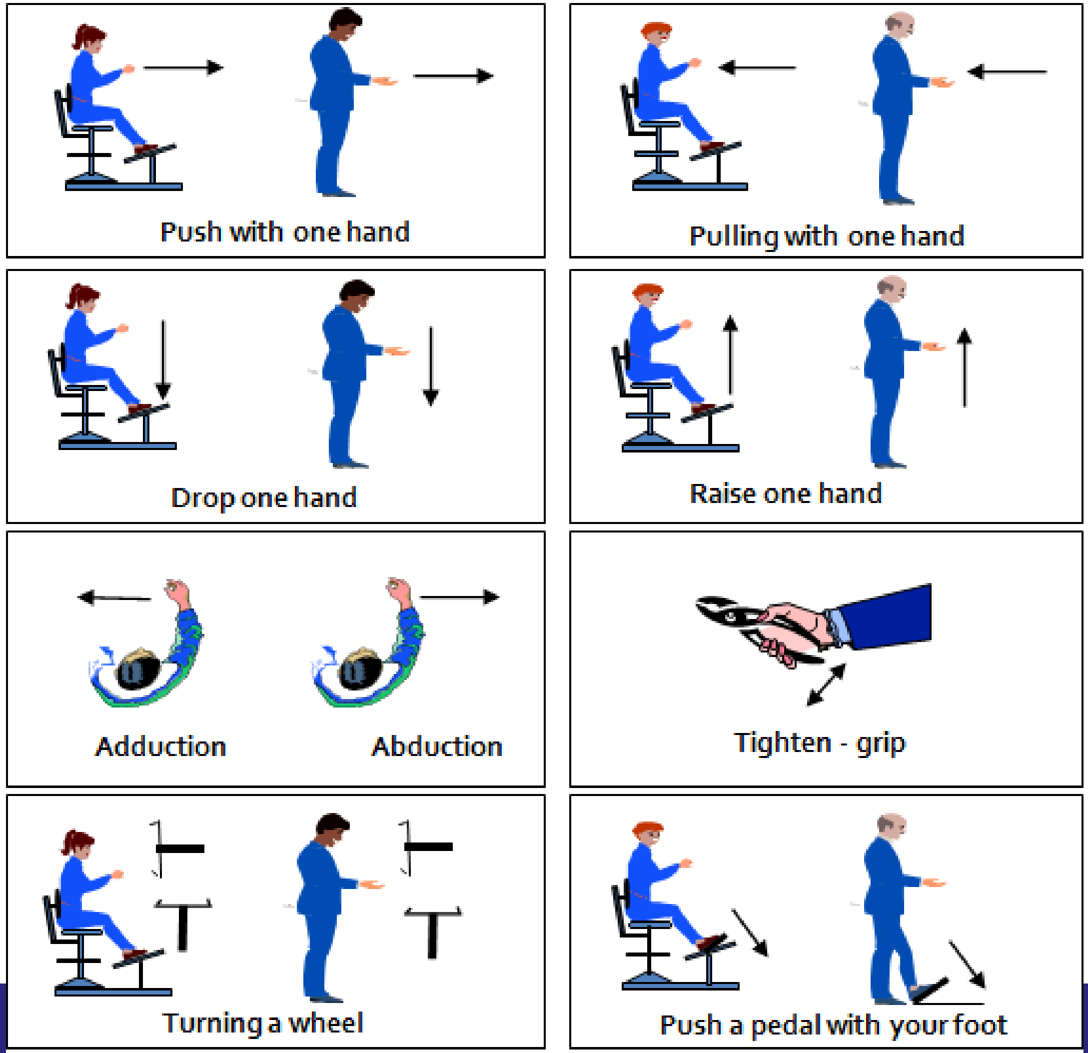
Principle 5. Reduce Excessive Motions > Efforts exerted

The physical workload depends on the forces exerted, the duration or frequency of maintaining these forces and the posture adopted to exert these forces. Forces occur when pushing, pulling or lifting an object.

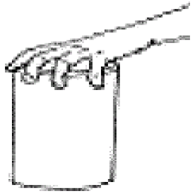
Limitation of effort

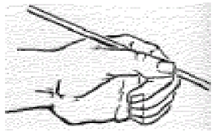
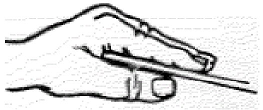


3.The ergonomic study

Principle 6. Minimize Contact Stress

- Avoid using the hand as a hammer (blows), clamps (efforts), press (prolonged pressure).
- Avoid maintaining prolonged static forces as well as forces applied with wrist rotation or extreme angulations.




Principle 7. Reduce Excessive Vibration

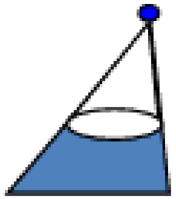
Excessive vibration causes pain to muscles, joints and internal organs; causes nausea and trauma to the hands, arms, feet and legs. Vibration is measured by its direction, acceleration and frequency on the body.

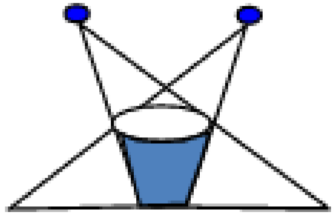
Recommended values without daylight

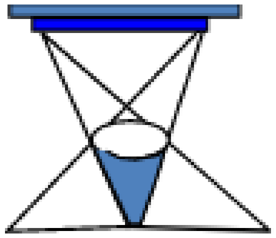
- Illumination level (offices) = 500 lux
- Illumination level (assembly of medium parts) = 500 lux
- Illumination level (assembly of small parts, control) = 750 lux
- Illumination level (assembly of very small parts) = 1500 to 2000 lux

Principle 8. Provide Adequate Lighting

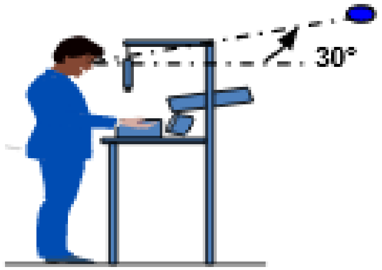
Influence on lighting homogeneity


A point source gives hard, sharp shadows

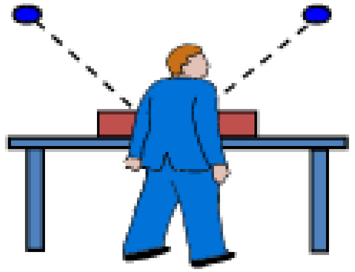

A second source attenuates some of the shadow cast by the first.

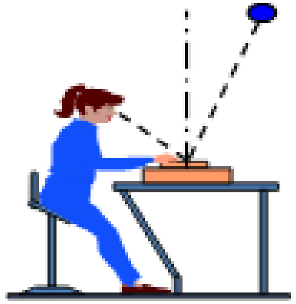

A large dimensional source softens the shadows

Glare suppression



The operator can only see the source if he raises his gaze more than 30°.


Good position of the sources removing both shadows and reflections


Position of the source to be avoided (risk of reflection)

Contrasts

Between the object and the immediate background

Good N

Medium N

Bad N

Too much lighting can degrade working conditions due to glare and / or unbalanced luminances

ECAM LaSalle

Other. Reduce noise

Recommended exposure values in workshops:
$L_{EX,8h*}$ < 80 dB (A) and/or Lpc < 135 dB (C)

Recommended noise level in offices:
Daily noise exposure L_{EXd} < 55 dB (A)

Recommended noise level on receipt at the machine or tool supplier:
Daily noise exposure: L_{EXd} < 70 dB (A)
Peak sound pressure: L_{PC} < 120 dB(C)

130 dB A	jet plane at 30 m
125 dB A	trap ball
120 dB A	pain threshold
115 dB A	hydraulic jackhammer, chainsaw
110 dB A	rock concert
105 dB A	circular saw blade
100 dB A	chainsaw, grinder
90 dB A	brushcutter, rotary printing press, sandblasting
85 dB A	driver's cab of some vans
80 dB A	lively street
70 dB A	analogue telephone ringing tone, TV set in normal operation
60 dB A	conversation at 1m
50 dB A	quiet office
40 dB A	quiet residential area
30 dB A	library
20 dB A	watch ticking

*The $L_{EX,8h}$ corresponds to the time-weighted average of the noise exposure levels for a nominal eight-hour working day, as defined by the international standard ISO 1999: 1990

3.The ergonomic study

Other. Reduce noise

$L_{EX,8h}$ | L_{PC}

Permissible limit value taking into account individual protection

87 dB(A) — 140 dB(C)

Noise Control Program
Obligation to wear personal protective equipment Signage of premises

85 dB(A) — 137 dB(C)

Reinforced medical surveillance - Information - staff training
Provision of individual protection

80 dB(A) — 135 dB (C)

Ambient noise levels (in dBA) allowing conversation to take place according to the distance between speakers.

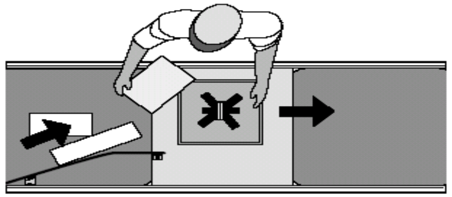
Distance between interlocutors	*0.35m*	*1m*	*2m*	*3m*
Very loud voice	*80*	*70*	*65*	*60*
Normal voice	*70*	*60*	*55*	*50*

The noise level must be adapted:
- to the exchange of information at the workstation
- to the complexity of the task (memorisation, information processing, etc.)

3.The ergonomic study

Choice of cycle time

Sensitive ZONE	Acceptable ZONE	Recommended ZONE	Acceptable ZONE
Risks of : - Non quality - TMS - De-productivity - Imbalance	Risks of MSD development To request the intervention of the competent authority concerned	Minimum risk	Risks of learning disabilities and impediments to versatility.



15 s
41 dmh

30 s
83 dmh

60 s
166 dmh

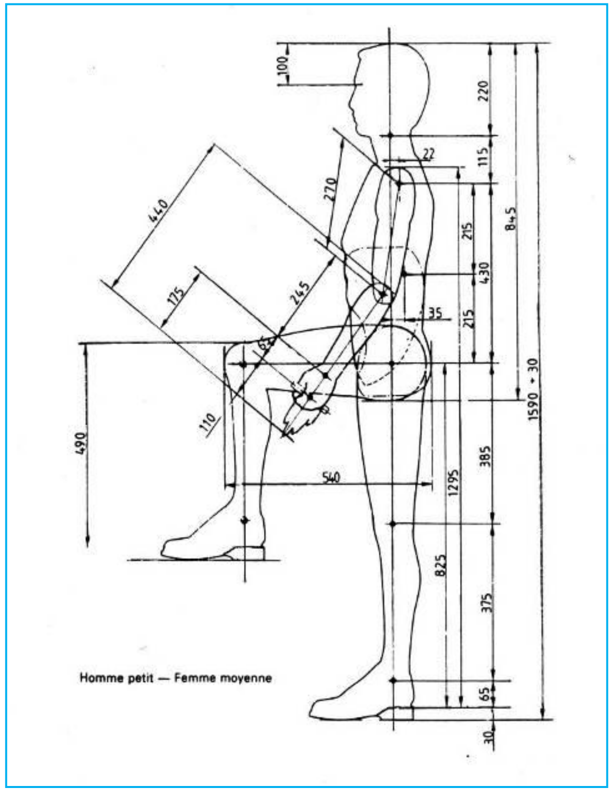
Cycle time

Carry out an in-depth analysis carried out by a working group made up of business specialists

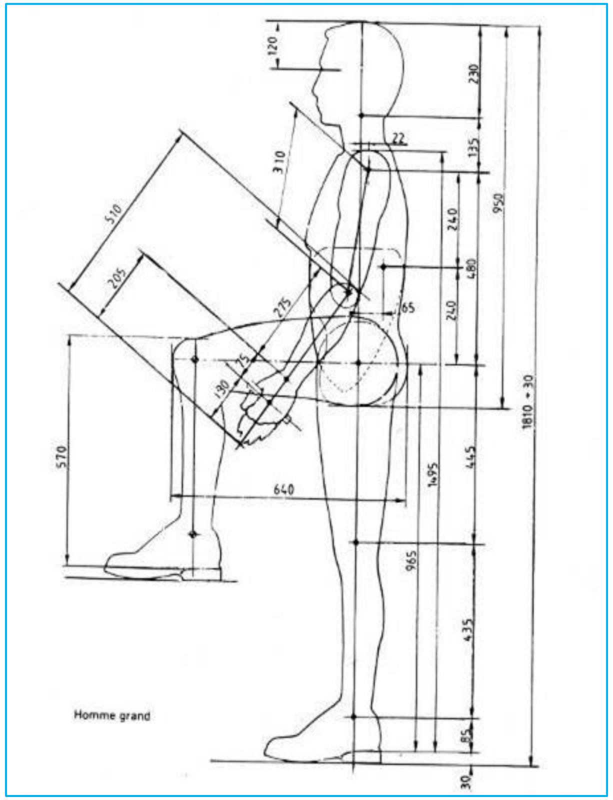
3.The ergonomic study

Anthropometric models

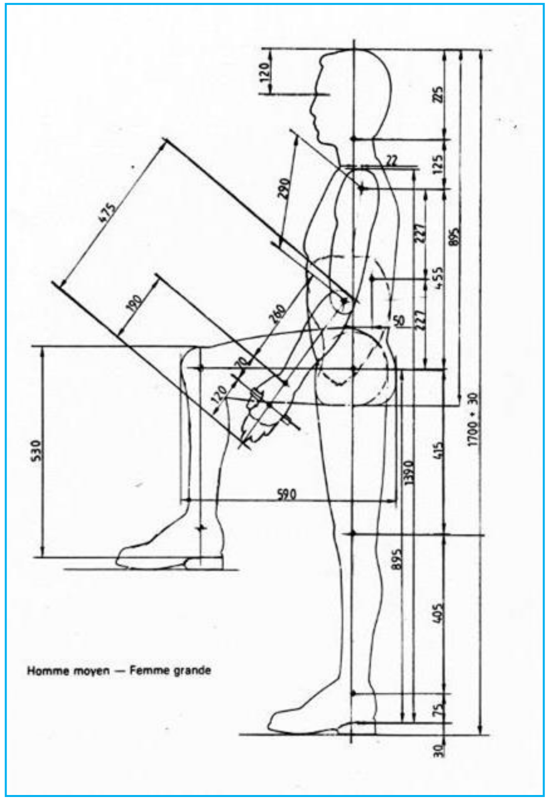
Small Man - Medium Woman



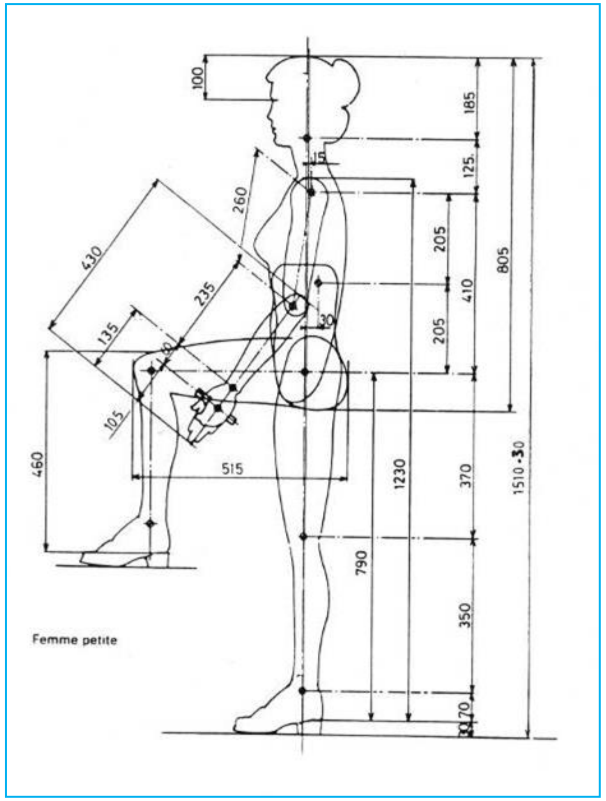
Tall Man



Medium Man - Tall Woman



Small Woman



Values selected for a particular country may not be appropriate for another geographical area depending on local populations.

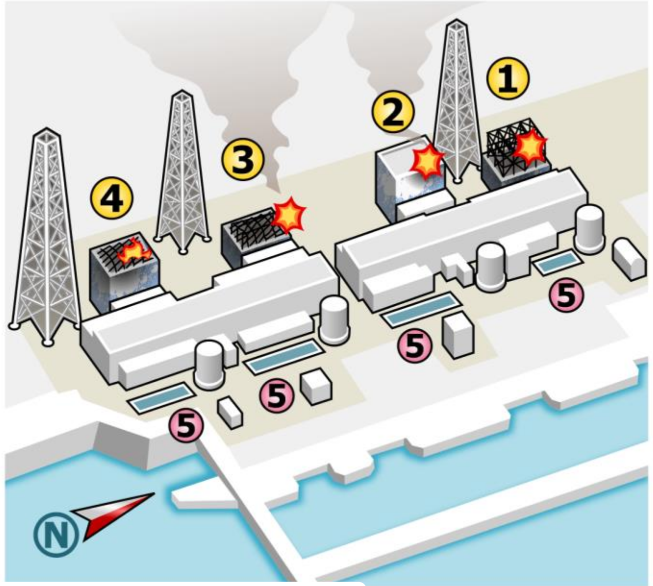
4. Failure Modes Effects and Criticality Analysis

Background story 2

March 11, 2011 Fukushima-Daiichi (JAPAN) disaster

A violent earthquake of magnitude 9, followed by a tsunami which strikes the Japanese coasts, cuts the electricity supply and damages the nuclear reactors of the power station causing one of the most serious nuclear catastrophe after Chernobyl.

This disaster is also the result of human error.



4.Failure Modes Effects and Criticality Analysis

The origin of this method dates back to the 1950s in the United States with the arrival and taxation of **Value Analysis*** in state markets.

Rapidly changing customer requirements and the growing importance of legal requirements, are forcing companies to apply rigorous techniques that identify and prevent potential problems.

The implementation in Europe only started from the 1980s but spread in all sectors of activity, and for all sizes of companies.

** Initially used to improve a product or service, value analysis is also used from design. Its purpose is to provide a 'product' with the functions it is asked to do, at the lowest cost.*

4.Failure Modes Effects and Criticality Analysis

Definition: **Analysis of Failure Modes, Their Effects and Criticality**

When the study only includes risk analysis and assessment.

- We are talking about FMEA (Failure Mode and Effects Analysis)

This study can be supplemented by a search for criticality.

- In this case, we are talking about FMECA Failure Mode, Effects and Criticality Analysis.

FMEA boils down to asking the following questions:

- ❑ **What will happen when a potential failure becomes reality?**
- ❑ **What are the consequences of this failure?**
- ❑ **What is the probability that this failure will appear?**
- ❑ **How can we detect and avoid this failure?**

4. Failure Modes Effects and Criticality Analysis

The aims of FMECA

- To identify the weaknesses of a system and to remedy them
- Specify the means to protect against certain failures
- To bring together the people concerned by a project
- Better understand the system
- Study the consequences of failure on reliability, availability, security and maintainability
- To classify failures according to certain criteria
- To facilitate conception
- To locate the critical operations likely to be at the origin of a non-conforming product

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4.Failure Modes Effects and Criticality Analysis

The different FMECA

FMEA can be used as a means of studying equipment life-long failures
If a forecast FMEA has been made, it is a good idea to check the hypotheses retained at the outset, in the light of lived experience.

The FMEA made during its lifetime, has objectives comparable to the forecast FMEA:

- Improvements
- Better adapted maintenance policy
- Better monitoring of operating parameters
- Faster fault diagnosis

Such an analysis also serves as a basis for discussion and reference in the adjustments made between manufacturer and user, and between the various departments concerned at the latter.
FMEA must also lead to the drawing of cause-effect diagrams which are easy to use in the search for the causes of failure.

Mean or machine FMEA: In this case we speak of **FMECA** (Analysis of Failure Modes, their Effects and their Criticality) to study rejects, breakdowns, touch-ups, production stoppages and optimize maintenance ranges.

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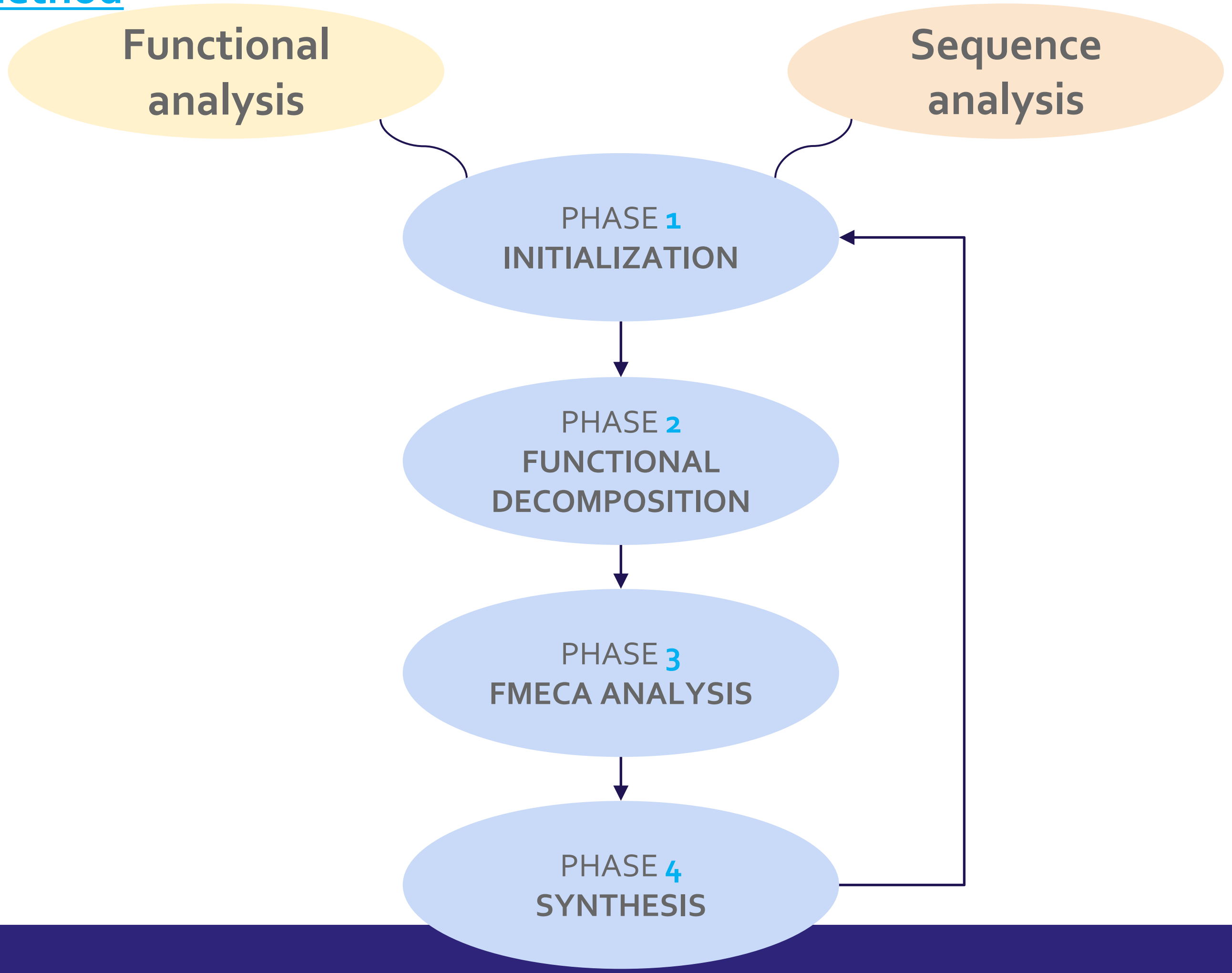
4.Failure Modes Effects and Criticality Analysis

The FMECA Machine method



4.Failure Modes Effects and Criticality Analysis

The FMECA Machine method

STEP 1: Review the process

- Use a process flowchart to identify each process component.
- List each process component in the FMEA table.
- If it starts feeling like the scope is too big, it probably is. This is a good time to break the Process Failure Mode and Effects Analysis into more manageable chunks.

STEP 2: Brainstorm potential failure modes

- Review existing documentation and data for clues about all of the ways each component can failure.
- The list should be exhaustive – it can be paired down and items can be combined after this initial list is generated.
- There will likely be several potential failures for each component.

STEP 3: List potential effects of each failure

- The effect is the impact the failure has on the end product or on subsequent steps in the process.
- There will likely be more than one effect for each failure.

STEP 4: Assign Severity rankings

- Based on the severity of the consequences of failure.

STEP 5: Assign Occurrence rankings

- Rate the severity of each effect using customized ranking scales as a guide.

STEP 6: Assign Detection rankings

- What are the chances the failure will be detected prior to it occuring.

STEP 7: Calculate the RPN

- Severity X Occurrence X Detection

STEP 8: Develop the action plan

- Decide which failures will be worked on based on the Risk Priority Numbers. Focus on the highest RPNs.
- Define who will do what by when.

STEP 9: Take action

- Implement the improvements identified by your Process Failure Mode and Effects Analysis team.

STEP 10: Calculate the resulting RPN

- Re-evaluate each of the potential failures once improvements have been made and determine the impact of the improvements.

4.Failure Modes Effects and Criticality Analysis

The FMECA table

It is the support for all of the previous steps which will make it possible to assess the level of criticality and to pilot corrective actions.

With this table the pilot of the project will be able to animate, pilot and communicate on the FMECA project.

	Failure Modes Effects and Criticality Analysis		FMECA MACHINE
	System:	Operating phase:	
	Sub-System:	Date of analysis:	Page:

Item number	Element	FUNCTION	FAILURE			Detection	CRITICITY				EVOLUTION						
			MODE	EFFECT	CAUSE		O	S	D	RPN	Corrective action	Resp.	Deadline	nO	nS	nD	New RPN

4.Failure Modes Effects and Criticality Analysis

The FMECA table

Criticity	Questions to ask
O: Occurrence	What is the probability of the failure mode resulting from a given cause? What is the corresponding frequency level?
S: Severity of failure effects	What is the level of severity corresponding to the identified effects of the failure?: • Equipment degradation? • Machine or line availability rate? • Non-conformity of the product produced? • Cost of maintenance? • Operator safety? • Environmental impact?
D: Probability of non-detection	The probability of non-detection depends on: • The existence of an anomaly that can be observed early enough • Detection actions implemented at the time of the study
RPN: Risk Priority Number	What is the resulting level for the criticality of the failure?

Failure Modes Effects and Criticality Analysis

FMECA MACHINE

System: Which machine is concerned by the study. *e.g.: Machining Center*

Operating phase: What is the operating phase studied e.g. Machining phase

Sub-System: What is the situation of the element (s) studied in the machine tree. *e.g. longitudinal movement table*

Date of analysis: 05/05/2017

Page: *1/1*

Item number	Element	FUNCTION	FAILURE			Detection	CRITICITY				EVOLUTION						
			MODE	EFFECT	CAUSE (5M + 5 Why)		O	S	D	RPN	Corrective action	Resp.	Deadline	nO	nS	nD	New RPN
Identify each failure with an identification number e.g. 1, 1.1, 1.2,2,etc.	 Precision ball bearing guide rail (SKF)	Infinitive verb + complement *e.g. Guide the table on the axis α*	1. Test all failure modes 2. What can happen to the element 3. How the element can no longer function properly *e.g. Blockage, Seizure, Rupture, Deformation, Wrong guidance, Fire,…*	1. List the most serious effects 2. What are the effects of the failure mode on the machine and / or its user *e.g. Material degradation, Performance loss, Breakdown, Production stop, Slowdown, Product non-compliance, Reject, Retouch, Direct maintenance cost, Personal injury, Pollution, Contamination*	1. List all the root causes 2. What can cause the failure mode on the machine *e.g. Under sizing Internal material fault, Routing, Defective adjustment, improper use, Lack of lubrication, Natural or premature wear, Shock, Overload, Tired*	1. List detections 2. What can warn of the failure mechanism *e.g. Part size, Part shape, Surface condition, material, Vibration, Heating, Lubricant degradation, Trace of wear, Abrasion, Corrosion, Crack, Scratch, Coloring, Appearance, Noise, Odor*	*Failure frequency*	*Severity of failure effects*	*Probability of non-detection*	*Risk Priority Number, Criticality = RPN = F×G×D*	*List proposals for corrective action*	*BMA*	30/05/17	…	…	..	…

ECAM Lyon	Failure Modes Effects and Criticality Analysis		FMECA MACHINE
	System: *Machining Center*	Operating phase: Machining phase	
	Sub-System: *Longitudinal movement table*	Date of analysis: 05/05/2017	Page: *1/1*

Item number	Element	FUNCTION	FAILURE			Detection	CRITICITY				EVOLUTION						
			MODE	EFFECT	CAUSE (5M + 5 Why)		O	S	D	RPN	Corrective action	Resp.	Deadline	nO	nS	nD	New RPN
1	Precision ball bearing guide rail (SKF)	*Guide the table on the axis α*	*Wrong orientation*	*Material degradation*	*Defective adjustment*	*Appearance*	*3*	2	3	18	*Add a poka-yoke*	*BMA*	30/05/17	*2*	*2*	*2*	*8*
2	Precision ball bearing guide rail (SKF)	*Guide the table on the axis α*	*Deformation*	*Direct maintenance cost*	*Lack of lubrication*	*Lubricant degradation*	*1*	2	2	*4*	*Update maintenance plan*	*BMA*	30/05/17	*1*	*1*	*2*	*2*
...																	

4.Failure Modes Effects and Criticality Analysis

Failure, their effects and causes

By failure, we mean quite simply the fact that a function is not or is poorly performed, that a component or a functional block does not function, functions poorly or does not meet the specifications or that an operation is not or poorly executed.

Generic failure modes:

1	Structural failure (failure)	16	Reduced flow
2	Physical blockage or jamming	17	Wrong start
3	Vibrations	18	Do not stop
4	Don't stay in position	19	Do not start
5	Does not open	20	Don't switch
6	Does not close	21	Premature operation
7	Failure in closed position	22	Operation after the scheduled time
8	Failure in open position	23	Incorrect entry (increase)
9	Internal leak	24	Wrong entry (decrease)
10	External leak	25	Wrong exit (increase)
11	Exceeds the upper tolerated limit	26	Wrong exit (decrease)
12	Unintended operation	27	Loss of entry
13	Intermittent operation	28	Loss of exit
14	Irregular operation	29	Short circuit
15	Wrong indication	30	Open circuit

4.Failure Modes Effects and Criticality Analysis

Failure, their effects and causes

To identify the effects, it is assumed that the failure actually occurred. It then suffices to define the consequence of the identified failure on the product on the means or the system for the user.

Examples:

- Unusable parts
- Destruction of another component
- Failure to respect capability
- Unable to handle
- Noise
- Degradation of appearance
- Leak
- Production stopped
- Security degradation

The search for causes consists in enumerating all the events likely to lead to the failure mode. As clear a description as possible facilitates the search and subsequent development of corrective actions.

Examples:

- Inadequate maintenance
- Insufficient lubrication
- Unsuitable means of production (machine capability too low)
- Incorrect preparation
- Corrosion before assembly
- Assembly error

In this context, one can carry out a systematic search on the basis of the cause-effect diagram

4.Failure Modes Effects and Criticality Analysis

Calculation of the RPN (Risk Priority Number)

Occurence - O

This step consists of estimating the probability of occurrence of the failure or non-compliance depending on the cause and assigning it a number of points.

Frequency or probability of occurrence	Occurrence ranking O	**Definition**
Very low frequency	1	Less than one failure per year
Low frequency	2	Less than one failure per quarter
Average frequency	3	Less than one failure per quarter
High frequency	4	Several failures per week

4.Failure Modes Effects and Criticality Analysis

Calculation of the RPN (Risk Priority Number)

Severity - S

It's a question of weighing the gravity of the failure according to the effect for the customer or the user.
The degree of severity can be increased depending on the consequences of the failure on the following operations.

Severity	Severity ranking S	Definition
Minor	1	• Production shutdown of less than 2 minutes and no significant degradation of the equipment
Significant	2	• Significant failure causing production to stop for 2 to 20 minutes or requiring overhaul or small repair on site • Product downgrading
Average	3	• Average failure causing a production stop of 20 min to 60 min or requiring a change of equipment, • Touch-up of product required or discarded
Major	4	• Significant failure causing production to stop for 1 to 2 hours or requiring major intervention on a sub-assembly • Production of non-conforming parts not detected
Catastrophic	5	• Serious failure, causing a shutdown of more than 2 hours or involving a heavy and costly intervention, personnel or environmental safety problems

4.Failure Modes Effects and Criticality Analysis

Calculation of the RPN (Risk Priority Number)

The probability of non-detection - D

This step consists of evaluating the probability of not having detected the cause of the fault or failure.
The higher the probability of detection, the lower the non-detection score.

Probability of non-detection	Detection ranking D	Definition
Obvious	1	Reliable detection of the cause of failure, clear sign of deterioration, automatic incident detection device (alarm)
Possible	2	There is a warning sign of the failure that is easily detectable but requires special action by the operator (visit, visual inspection, etc.)
Unlikely	3	The warning sign of the failure is not easily detectable, not very exploitable or requires action or complex means (disassembly, equipment,etc.)
Impossible	4	There are no early signs of failure

4.Failure Modes Effects and Criticality Analysis

Calculation of the RPN (Risk Priority Number)

Summary and Actions to be taken depending on the level of criticality

RPN (O x S x D)	Example of corrective action to be taken
1 < RPN < 12 Negligible Criticality	No design changes, corrective maintenance
12 ≤ RPN < 16 Average Criticality	Element Performance Improvement, Systematic Preventive Maintenance
16 ≤ RPN <20 High Criticality	Revision of the design of the sub-assembly and the choice of components, special surveillance, conditional preventive maintenance
20 ≤ RPN < 80 Forbidden Criticism	Complete questioning of the design

Thank you
for your attention

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